Predictive Model Plan – Student Template

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1. Model Logic (Generated with GenAI)

Model Type: Random Forest Classifier

Objective: To predict the binary outcome of Delinquent_Account (1 for at-risk, 0 for stable) by analyzing historical behavior and financial health metrics.

Key Components & Workflow:

1. **Data Ingestion & Pre-processing:** The model ingest historical customer data. Categorical variables like Employment_Status and Month_1 through Month_6 are transformed using One-Hot Encoding, while numerical features are scaled to ensure consistent weighting.
2. **Feature Ensemble:** The Random Forest algorithm constructs multiple decision trees during training. Each tree is built using a random subset of features and data points (bagging), which reduces the risk of the model "overfitting" to specific anomalies in the Geldium dataset.
3. **Decision Path:** Each tree makes an individual prediction based on thresholds. For example: *Is Credit Utilization > 0.8? → Is Missed Payments > 2? → Predict Delinquent.*
4. **Probability Aggregation:** The final output is determined by the "majority vote" across all trees. The model provides a probability score (e.g., 0.85) rather than just a simple "Yes/No," allowing the Collections team to prioritize customers with the highest risk scores.

Action: Model Summary

The chosen model is a **Random Forest Classifier**, an ensemble learning method that combines multiple decision trees to produce a highly accurate and stable prediction. It is designed to capture complex, non-linear interactions between a customer's financial status and their likelihood of delinquency, providing the Collections team with a prioritized "Risk Score" for every account.

Top 5 Input Features:

1. **Credit_Utilization:** The most direct indicator of liquidity stress and the strongest predictor of impending delinquency.
2. **Missed_Payments:** A historical count that reflects a customer's immediate repayment reliability.
3. **Debt_to_Income_Ratio (DTI):** Measures the customer's total financial burden relative to their capacity to pay.
4. **Month_6 (Most Recent Status):** The most recent payment behavior (Paid, Late, or Missed) serves as a critical "real-time" indicator of a downward trend.

5. **Credit_Score:** A foundational metric of creditworthiness that provides long-term context for current financial behaviors.

2. Justification for Model Choice

The Random Forest Classifier was selected because it offers an optimal balance between predictive precision and operational transparency, which is critical for Geldium Finance's regulatory compliance and customer trust. Unlike basic linear models, Random Forest is highly effective at identifying the complex, non-linear relationships—such as the sudden risk spikes associated with high credit utilization—that were uncovered during the initial data analysis. This enhanced accuracy allows the Collections team to move from broad historical segments to a high-precision "Early Warning System," specifically targeting high-risk accounts before a delinquency event occurs. While being a sophisticated ensemble method, it remains relatively easy to implement and provides "Feature Importance" metrics, ensuring that every intervention decision can be explained and justified to stakeholders. Ultimately, this model fits Geldium's need for a robust, scalable tool that improves financial outcomes while maintaining the explainability required for responsible lending.

3. Evaluation Strategy

To ensure the Random Forest model is reliable and responsible, we will use a multi-faceted evaluation framework that prioritizes the cost of missing a delinquent customer against the risk of alienating a good one.

Key Performance Metrics:

- **Recall (Primary Metric):** This measures the model's ability to find all actual delinquent customers. For Geldium, a high recall is critical because the cost of "missing" a customer who eventually defaults is much higher than the cost of a proactive check-in call.
- **Precision:** This measures how many of the customers flagged as "High Risk" actually became delinquent. High precision ensures that the Collections team is not wasting resources on customers who are performing well.
- **F1-Score:** The harmonic mean of Precision and Recall, used to ensure we have a healthy balance between the two.
- **ROC-AUC (Area Under the Curve):** This provides an aggregate measure of performance across all possible classification thresholds. An AUC of 0.85 or higher would indicate excellent discriminatory power.

Interpretation of Metrics: A **high Recall** with **moderate Precision** is the target "sweet spot" for Geldium. If we achieve 90% Recall, it means we caught 9 out of

10 potentially delinquent accounts. If Precision is lower (e.g., 60%), it means some "Low Risk" customers were contacted, but in a proactive environment, this can be framed as a "Customer Wellness Check" rather than a debt collection call, mitigating the downside.

Bias Detection and Reduction: To ensure fairness, we will perform **Disparate Impact Analysis**. This involves:

- **Sub-group Testing:** Comparing the model's error rates across different demographics (e.g., Age, Location, and Employment Status) to ensure the model doesn't unfairly "over-predict" delinquency for specific groups.
- **Feature Importance Audit:** Using SHAP values to confirm that the model is making decisions based on financial behavior (like Utilization) rather than proxy variables that might introduce systemic bias.

Ethical Considerations:

- **Transparency:** Every customer flagged as "High Risk" must have an explainable reason (e.g., "Utilization increased by 30%"). We will avoid "Black Box" decisions.
- **Proactive vs. Punitive:** The model results will be used to offer **assistance** (like restructured payment plans) rather than immediate credit limit cuts or negative bureau reporting.
- **Data Privacy:** We will ensure that the model utilizes only the minimum necessary data points and that customer information remains encrypted and accessible only to authorized personnel.